

FishAI v1.5: A Bit-Budget Memory Ladder as an Honest Difficulty Axis

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Abstract

Until now the FishAI engine’s easy tier was a six-event memory window plus a 25% uniform blunder rate — a difficulty knob that is neither monotone, interpretable, nor human-shaped. FishAI v1.5 replaces it with the bounded-memory model of Sanjay Kannan’s Literature framework: memory capacity capped *in bits* under a fixed fact tariff (2 bits per card fact, 1 per basis fact), with the eviction policy the strategy corpus recommends — rank facts by contestability, not recency — implemented as a stateless, pure re-derivation from the full public log at every decision. The design is measured under the project’s pre-registration discipline, eight predictions with verdict rules fixed before each run. The ladder is confirmed monotone across all ten measured budgets (P1: every adjacent rung non-negative, the sharpest at $z = 46.8$; P2: the ∞ cell an exact mirror over 3,000 duplicate pairs). Calibrating the shipped tiers onto the ladder (P3, mixed): medium is worth 32.2 bits [30.8, 35.1]; the old noise-based easy tier measures *below the zero-bit floor* (0.0446 set-share against the floor’s 0.1313 — a memoryless bot that reasons beats a remembering bot that dices); hard sits above every finite rung. Evidence-age analysis (P4–P6, all mixed as printed) finds the S45 signature where the model predicts it — budgets up to 48 bits decay, with half-lives 5/13/17/33/49 events, rising in bits where defined — while the noise tier is cliff-edged at its window ($z = 258.9$) yet not flat inside it, and a composition effect makes the pooled and seed-clustered estimators disagree in sign for the full-memory policies. The headline refutation is P7: memory pressure was predicted to erode the behavioural style signature, and instead the v1.0 classifier reads bounded ecologies *better* — top-1 rises from 15.4% at 16 bits to 23.0% at 64 bits against 22.4% at ∞ , and the violating rung (-0.0053 ± 0.0019) survives a Bonferroni $\times 3$ annotation at $p = 0.00885$. A single-seat follow-up (E4b), registered after review, isolates the read seat: its 50-seed pilot confirmed a flat curve but at 52% power — its top rung’s confidence interval contained the entire P7 effect — and the pilot’s commit overclaimed; the correction is on the record, and the 300-seed run of record (99.97% post-hoc power at the P7 effect size) confirms P8 flat at matched read count. Within the designs measured, the P7 shift is a property of the bounded ecology, not of the bounded seat’s own end-of-game signature; the cross-design difference ($z = 2.11$) is reported as exactly that and enters no verdict.

1 Introduction

Every difficulty system makes a claim about what a weaker player *is*. The FishAI engine’s shipped answer, inherited from v0.5 [4], was the standard one: an easy bot remembers only the last six public events, skips constraint inference, and replaces 25% of its asks with random legal ones. That answer is expedient and wrong in three measurable ways. It is not monotone — noise interacts with position, so “more noise” is not reliably “weaker” in any calibrated sense. It is not interpretable — there is no single number a player can reason about. And it is not human-shaped: humans do not

play well and then dice-roll into blunders; they *forget*, oldest and least relevant facts first, and their errors inherit that structure [8, 2].

The project’s research synthesis catalogues an alternative with all three properties [8]: the bounded-memory player of Sanjay Kannan’s Literature framework (S44 in the synthesis’s numbering), in which an agent’s memory is capped in *bits* under a fixed tariff per atomic fact, and the interesting decision — given k bits, what should you remember? — is delegated to an explicit eviction policy. The synthesis pairs it with two neighbours that complete the design: the recency player (S45), whose observable signature is a decay of effectiveness with the age of the supporting evidence, and the triage memoriser (S47), the human-facing name for the eviction policy the strategy corpus actually recommends — memorise the books you can contest, and flush what is resolved.

FishAI v1.5 implements that model as a first-class policy arm of the deterministic **us54** engine (Section 5), and then measures it under the discipline the project’s adaptive suite established [5]: pre-registered predictions with verdict rules fixed before each run, duplicate deals, paired and seed-clustered standard errors, health gates that void a run rather than qualify it, and one committed artifact [14] from which every number in this paper is read. Four experiments and two registered follow-ups answer four questions. Is the ladder monotone? (Yes, at every rung.) Where do the shipped tiers land on it? (Medium at 32.2 bits; the old easy tier below the ladder’s own floor.) Are bounded bots human-shaped in the S45 sense? (Where the model has room to bind, yes, with half-lives that rise in bits; the full-memory verdicts are mixed for reasons worth stating exactly.) And does memory pressure erode a seat’s behavioural style signature, as the v1.0 observation work [6] would suggest? Here the pre-registered prediction was **refuted**, and the refutation is this paper’s headline: under mild memory pressure, 64-bit ecologies are *more* classifiable than full-strength ones, and a registered single-seat follow-up — including a pilot whose conclusion had to be corrected on the record, a sequence Section 6.4 reports in full — localises the effect in the ecology rather than in the read seat’s own signature, within the designs measured.

2 Related work

The primary source is Kannan’s Literature framework and its reference implementation, both read in full by the project’s research synthesis [3, 8]: the framework specifies an agent “constrained by imperfect memory” via a bit parameter and a per-fact price list, and its reference **ActivePlayer** implements ranked retention with a **spotlight()** focus-book mechanism — the direct ancestors of Section 5. The idea that agents are resource-bounded reasoners rather than noisy optimal ones goes back to Simon [1]; the specific empirical shape assumed here — retrieval success decaying with the age of the supporting evidence, at rates that track the environment — is the memory literature’s [2], and the S45 entry of the synthesis [8] argues the decay curve is the single best skill estimator recoverable from a Literature log. Within the project, this paper leans on the v0.5 roster and measurement discipline [4], the v1.0 classifier whose accuracy harness E4 reuses [5, 6, 12], and the inert-axis result that bounds what any public-log observer can see of this roster [7, 10].

3 Background

Literature (Canadian Fish) under the **us54** dialect [9, 4]: 54 cards in nine half-suits (“books”) of six, six players in two teams, public asks gated by a holding licence, out-of-turn declares in a deterministic window, any declaration error awarded to the opponents, race to five sets, ties impossible. The engine is pure, deterministic TypeScript; one seed reproduces a byte-identical game [15].

Three prior measurements matter here. First, the v0.5 roster: nine parameterised styles over one shared full-strength inference engine, with Balanced the tuned control [4]. Second, the difficulty tiers as shipped: **easy** is a six-event log window without set constraints, a 25% uniform ask-blunder rate, and no claim planner; **medium** runs the full knowledge engine; **hard** adds the constraint-refined hit probability — the skill at which every roster style, and the bounded arm below, plays [15]. Third, the v1.0 observation layer: fourteen public-log features, a calibrated diagonal-Gaussian classifier, and a measured end-of-game top-1 of 22.4% over nine styles against an 11.1% chance floor [6, 13]. Experiment E4 runs that exact instrument over bounded play.

4 The S44 model and the cost tariff

The model prices memory, not behaviour. An agent holds at most k bits of derived facts, where the tariff — fixed by the framework, not tuned by this project [3, 8] — is:

Fact kind	Reading	Cost
has-card	player X has card Z	2 bits
lacks-card	player X does not have card Z	2 bits
basis	player X holds at least one card of book Y	1 bit
no-basis	player X holds no card of book Y	1 bit

The framework’s specification names the first three prices; its reference implementation’s fact tuples carry the negative basis fact as a fourth kind, priced like the positive [3, 8]. Two properties of the tariff do real work. A basis fact costs half a card fact and is often worth more — it is what gates ask legality and drives hit priors — so a good eviction policy is not a compression problem but a valuation problem. And the framework does no enforcement: the agent gets **self.limit** bits, tracks **self.free**, and the discipline is its own; the reference **ActivePlayer** *refills memory from scratch each turn*, re-deriving every fact and re-spending the budget down a relevance ranking anchored on a focus book chosen by its **spotlight()** routine. That refill semantics — not an incremental store with evictions — is what Section 5 implements faithfully.

The synthesis’s calibration argument [8] sets the scale: worst-case perfect recall of card facts alone runs to 2 bits per card per seat — hundreds of bits (the framework’s own constants are for its 52-card, seven-card-major deck; the model transfers to **us54**, the constants do not). A two-digit budget is therefore a genuine handicap, and Section 7.1 measures that on these deals the axis saturates by 128 bits.

Why this replaces noise as the difficulty knob is the synthesis’s argument, which the experiments test rather than assume: a bit budget is *monotone* (more bits can only add facts; P1), *interpretable* (one number a player can reason about), *human-shaped* (a relevance-ranked bounded memory forgets the oldest, least-contestable facts, so the S45 decay curve arises by construction; P5), and *composable* (a bounded Hoarder is still a Hoarder — the budget caps the memory, never the style or the reasoning; E4 exercises exactly this).

5 The engine: a stateless bounded-memory arm

The v1.5 arm is a fifth **PolicySpec** shape, {**bounded: true, bits, style**} — budget in bits, roster style to play over the restricted knowledge (default **Balanced**) — resolved inside **decide** exactly as the adaptive arm is [5, 15]. Everything below is a pure function of (view, policy, seed): no **Date**, no **Math.random**, no state between calls. The engine’s own summary of the division of labour:

the budget caps the memory, never the reasoning — the chosen roster style runs at hard skill over whatever knowledge survives the budget [15].

5.1 Derivation, then reconstruction

The implementation is two passes over public information [15]. **Derivation:** the full-fidelity recorded walk of the knowledge engine — the identical ingestion `buildKnowledge` runs at hard skill — read back as timestamped atomic facts of the four tariff kinds. Nothing is invented: a hit locates a card; an ask certifies the asker’s basis and their lack of the named card; a miss certifies the target’s lack; historical count exhaustion certifies eliminations; a claim resolves its book and retires every fact on it — exactly the certifications the observation layer documents for this log [6]. **Reconstruction:** the *kept* facts are replayed in log order onto a fresh working state through the same primitives the knowledge engine uses, then finished by the shared `finishKnowledge` step — own-hand injection, fixpoint propagation over the *current* public counts, materialisation. The output is the same `Knowledge` shape every style policy consumes, so all nine roster styles run over it unchanged.

Because the reconstruction reuses the knowledge engine’s own machinery, the **large-budget equivalence anchor** holds by construction rather than by a parallel implementation staying in step: with every fact kept, the replay reproduces the full walk’s state, and the arm’s decisions are identical to the unbounded style’s over whole games — pinned by test, and re-verified at run scale by the ∞ cells of Sections 7.1 and 7.5 [15, 14].

5.2 What is free, and what never enters the pool

Own hand is free — the seat can see it, so the injection runs at every budget, including zero. The public board state is free — resolved books are seeded gone from the view’s book state, and current counts drive the final propagation — because the synthesis’s S48 entry keeps *observation history* and *display policy* as separate axes: what the table shows is not what the agent remembers [8]. The S45 subtlety identified for the old log-window tier applies with force here: a fact-filtered replay of running hand counts would be wrong the moment a hit’s fact was evicted, so the reconstruction never replays counts at all — count exhaustion runs only inside the full-fidelity derivation, and its conclusions reach the replay as kept lacks-card facts [8, 15].

Facts the free state reproduces at any budget are never enumerated, so the budget is never spent on them: everything on a resolved book; the viewer’s own lacks and held cards; the viewer’s own basis constraints. Subsumption keeps the accounting honest — a card fact subsumes the eliminations on its card, and a no-basis fact compresses up to twelve bits of per-card lacks into one [15].

5.3 Spotlight ranking and the budget

The eviction policy is S47’s, under the reference implementation’s mechanism: each live book is scored by how much the seat’s own hand plus the derived facts bear on it (2 per held card, plus the bit cost of every pool fact on the book); facts rank by book score descending, then recency descending, then a stable total order (card in deck order, seat, kind); the kept set is the longest ranked prefix whose total cost fits the budget [15, 8]. Contestability leads and recency tiebreaks — the corpus’s advice, not this project’s invention: save your memory for the half-suits you can still contest, and flush what is resolved [8]. At `bits`: 0 nothing is kept and the policy sees own hand plus public board only — the synthesis’s own-hand-only regime (S46), and still never an illegal move; the test suite pins the zero-budget floor, the budget accounting (kept cost never exceeds bits; the kept set is a ranking prefix; ties deterministic), `decide/decideExplained` parity for the new arm,

and purity [15]. The explanation trace names the budget (“ k -bit memory”), so downstream surfaces cannot misrepresent what played.

6 Experimental design and pre-registration

6.1 Shared discipline

All experiments inherit the adaptive suite’s discipline [5, 11]: duplicate deals (each seeded deal played in both orientations, start seat rotating with the pair index), per-seed paired estimators for adjacent comparisons, seed-clustered standard errors wherever compositions share seeds, and health gates — 0 illegal actions, 0 capped games, 0 invariant violations, and under `us54` 0 ties, 0 voids, 0 non-clinch endings, with `distinctSeeds` equal to the expected count in every cell — that *void* a run rather than footnote it. Decisions are seeded `hashSeed('${seed}:${moveIndex}')()`; the runner is a pure function of its task (the worker pool is a throughput device, not a source of variation); an FNV digest of the records travels in the artifact; and the committed artifact `src/lab/data/bounded-results.json` sits behind a boundary validator mirroring the adaptive one [14, 15]. The base suite is 85,200 games in 308 seconds of wall clock; the E4b pilot and power runs add 7,200 and 43,200 games [14]. Every number in this paper is a field of that artifact (or of the committed v1.0 artifact where the baseline is cited [13]), quoted at the artifact’s own precision.

6.2 The experiments

1. **E1 — ladder monotonicity.** A team of three bounded-Balanced seats against an unbounded Balanced team at hard skill — the same full-strength engine the budget caps — at ten budgets, bits $\in \{0, 8, 16, 24, 32, 48, 64, 96, 128, \infty\}$ (∞ implemented as 10^6 , provably above the derivable pool), 3,000 duplicate pairs per rung on *one* shared seed list (prefix `bounded-v1`), so every budget replays identical deals and adjacent rungs compare per seed. The metric is mean set-share, $\text{sets}_A / (\text{sets}_A + \text{sets}_B)$, not win rate: the ladder needs a dial that keeps moving after the win rate saturates.
2. **E2 — tier calibration.** The three shipped tiers against the same reference opposition, on the head of the E1 seed list, so each tier’s share sits on the E1 curve per-deal and its bits-equivalent is read off by interpolation.
3. **E3 — evidence age.** No games of its own and no engine instrumentation: computed post-hoc from the E1/E2 records, which retain the full public log. The estimator, fixed in the registration: at ask event i , every card whose public location a hit established at event j (unmoved and unretired since) that the acting seat could legally ask of the correct holder is one availability observation at age $i - j$; the seat *exploited* it iff its actual ask was exactly that card at that holder; the hit column is the actual ask’s outcome. Hands are reconstructed from the seed’s deal plus the log’s hits and claims, so ask legality is decided from ground truth. Clustered comparisons cluster by seed; the young band is ages 1–8, the old band 33+; a half-life is the lower edge of the first band with at least 200 observations whose exploit rate is at or below half the youngest band’s — a curve that never falls that far has none.
4. **E4 — style under memory pressure.** The v1.0 classifier-accuracy harness [5, 6], budget-capped: for each bits $\in \{16, 32, 64, \infty\}$, all 36 unordered style pairings play 50 single games with *both* teams bounded at that budget, on the v1.0 accuracy seed list (`clsacc-v1`); the classifier reads every seat once at end of game against the committed full-strength fingerprints

[12] — 1,800 games and 10,800 read seats per cell. At ∞ the arm is decision-identical to the bare styles by the anchor pin, so that cell is a re-run of the v1.0 accuracy experiment and must reproduce the committed 22.4% exactly.

5. **E4b — single-seat attribution** (registered after the Phase 2 review; Section 6.4). E4’s grid, seeds and pairings, but only one seat is bounded: the *read seat*, seat $2 \cdot (\text{game mod } 3)$ — the three team-0 seats in rotation — always playing the pairing’s team-0 style under the budget while the other five seats play their bare full-strength styles exactly as the v1.0 harness seats them. The start seat rotates game mod 6, so the read seat occupies every relative table position uniformly. P8 scores the read seat only; truth is always the pairing’s first style, so the triangular pairing scheme gives by-style read counts that fall with roster position, identically at every budget, and the last roster style is never the read truth. The ∞ cell carries an in-run health gate: every ∞ -budget game is replayed with all six seats bare and required to match event-for-event, read-for-read and step-for-step (replays are checks, not records). The design holds the ecology — opponents, partners, game length — approximately at the full-strength distribution the fingerprints were calibrated on, isolating the read seat’s own signature.
6. **E4b-power — the P8 run of record** (registered after the E4b review; Section 6.4). The same design at 300 seeds per pairing — 10,800 reads per cell, matching P7’s read count — on a fresh seed prefix (`clsacc-power-v1`) disjoint from the pilot’s; mapping, grid, estimator and rule unchanged.

6.3 Predictions and verdict rules

Eight predictions travel verbatim inside the artifact [14]; they are typeset here with mathematical symbols but otherwise as registered. P1–P7 were registered 2026-08-29, before any Phase 2 run; P8 on 2026-08-30, before any E4b run.

- P1** E1 (ladder monotonicity): set-share is non-decreasing in bits, within $2 \cdot (\text{paired SE})$ at every adjacent pair.
- P2** E1 (health, not prediction): at ∞ the pairing is an exact mirror; duplicate-deal set-share is 0.5000 exactly by construction. A deviation is a harness bug, full stop.
- P3** E2 (question, direction only): each shipped tier lands at a finite, orderable bits-equivalent (easy < medium < hard). No numeric prediction — this is the measurement the ladder exists to provide.
- P4** E3: full-memory policies are age-flat (the S48 log-reader regime, by construction).
- P5** E3: bounded policies decay with age, and the decay half-life increases with bits (the S45 signature, by construction of spotlight + recency eviction).
- P6** E3: the old easy tier (`logWindow 6 + 25%` uniform noise) is age-flat inside its window and cliff-edged at it — i.e. noise is not human-shaped; the paper’s motivating contrast.
- P7** E4 (direction): top-1 accuracy is non-increasing as bits shrink — memory pressure erodes the behavioural signature. Where it lands relative to the 22.4% full-strength baseline is the measurement, not a prediction.

P8 E4b (single-seat attribution): top-1 accuracy at the bounded read seat is non-increasing as its bits shrink, by the same adjacent-rung 2·SE rule as P7 (any violating rung refutes). The ∞ cell must reproduce the corresponding full-strength read exactly (health, not prediction).

The verdict rules were fixed with the predictions and ship in the code that assembles the artifact [15], all on the adaptive suite’s 2·SE discipline. P1, P7, P8: every adjacent rung must satisfy $\Delta \geq -2 \cdot \text{SE}$ (per-seed paired); any violation refutes. P2: integer-exact mirroring of every ∞ pair; a deviation refutes and independently voids the run. P3: confirmed iff all three tiers are finite and strictly ordered; ordered-but-not-all-finite is mixed. P4: a full-memory policy is age-flat iff $|\text{young} - \text{old}| \leq 2 \cdot \text{SE}$ (seed-clustered); both of the two full-memory policies flat confirms, one is mixed. P5: every finite budget must decay significantly *and* have a defined half-life, non-decreasing in bits with at least one strict increase; anything between “all hold” and “none decays” is mixed. P6: the old easy tier must be flat inside its window ($|\text{rate}(1-3) - \text{rate}(4-6)| \leq 2 \cdot \text{SE}$) *and* cliff-edged at it; one of two is mixed. A refuted prediction is emitted as **refuted**, not massaged.

6.4 The registration timeline, and a correction of record

The sequence matters more than usual in this paper, because it includes a correction, and the correction is part of the method rather than a blemish on it.

1. **2026-08-29:** E1–E4 designs and P1–P7 registered; the base suite runs; P7 is refuted (Section 7.4).
2. **Phase 2 review:** the review confirms that E4’s both-teams design *cannot* attribute the P7 refutation — bounding both teams changes the whole ecology (opponents, partners, game length) along with the read seat’s signature, and the classifier’s fingerprints were calibrated on full-strength play. The review also recommends a multiplicity annotation: Bonferroni $\times 3$ over each adjacent-rung family, reported *alongside* the committed verdicts, which are not changed.
3. **2026-08-30, before any E4b run:** E4b registered — the single-seat design, P8, and the annotation; the read-seat mapping was chosen by the implementer and documented in the run’s `meta.notes` before running, as the registration required.
4. **The pilot and its overclaim:** the 50-seed E4b pilot runs; P8 comes out confirmed (flat). The commit recording it claimed the P7 shift was “not of the read seat’s signature”. That claim was too strong: what a flat curve at 1,800 reads per cell licenses is only that the effect *does not appear in the single-seat design at that read count*.
5. **The E4b review:** computes the pilot’s power at exactly the P7 effect size: about 52% (minimum detectable effect 0.0051 against the P7 effect 0.0053; the pilot’s $64 \rightarrow \infty$ interval $[-0.0067, +0.0034]$ contains the entire P7 effect). The pilot’s CONFIRMED is therefore an underpowered null.
6. **2026-08-30, after that review and before any run:** E4b-power registered. Seeds $50 \rightarrow 300$ (reads $1,800 \rightarrow 10,800$ per cell, matching P7’s read count; $\text{MDE} \approx 0.0021$ under the same rule), fresh disjoint seed prefix, everything else unchanged. The 300-seed run becomes the P8 verdict of record; the pilot is retained verbatim in the artifact, clearly labelled; both are reported whatever they say. The registration notes the direction of the risk honestly: raising N after a null could only have refuted the project’s own headline attribution — which is the direction honesty points. A correction commit states the pilot conclusion’s within-design limit; no history is rewritten.

Table 1: E1, the ladder: three bounded-Balanced seats vs. an unbounded Balanced team at hard skill, 3,000 duplicate pairs per rung on one shared seed list [14]. Δ is the per-seed paired delta from the previous rung; the pre-registered rule is $\Delta \geq -2 \cdot \text{SE}$ at every rung.

Bits	Set-share	SE	Mean moves	Δ from prev.	z
0	0.1313	0.0017	798.2	—	—
8	0.2449	0.0022	768.9	$+0.1135 \pm 0.0028$	41.11
16	0.3921	0.0026	741.1	$+0.1472 \pm 0.0031$	46.76
24	0.4361	0.0023	720.5	$+0.0440 \pm 0.0030$	14.50
32	0.4653	0.0020	704.5	$+0.0292 \pm 0.0025$	11.49
48	0.4909	0.0011	688.5	$+0.0257 \pm 0.0020$	12.52
64	0.4977	0.0006	685.1	$+0.0068 \pm 0.0011$	6.04
96	0.4999	0.0001	683.3	$+0.0022 \pm 0.0006$	3.80
128	0.5000	0	683.2	$+0.0001 \pm 0.0001$	0.90
∞	0.5000	0	683.2	$+0.0000 \pm 0.0000$	0.00

7. **Guard truthfulness:** the same review’s integrity findings are fixed before the power run — the artifact-extension guards’ prose narrowed to what they actually check, the registered prediction texts authenticated against the code’s own copies, the base artifact pinned by records digest, and the analysis step made to fail (not note-and-proceed) when the records file is absent [15].

7 Results

Health first, because a run is void without it: across all three runs (base, pilot, power) the counters are clean — `illegalActions`, `cappedGames`, `invariantViolations`, `ties`, `voids` and `nonClinch` all zero, and `distinctSeeds` equal to the expected count in every cell [14].

7.1 P1 and P2: the ladder is monotone, and the anchor holds

Table 1 is the axis. Set-share rises from 0.1313 ± 0.0017 at zero bits to exactly 0.5 at the top, and every one of the nine adjacent deltas is non-negative — the first seven each individually significant (the smallest at $z = 3.80$), the two top rungs statistically zero. **P1 is confirmed:** 0 of 9 rungs violate $\Delta \geq -2 \cdot \text{SE}$; the most negative rung is $128 \rightarrow \infty$ at $z = 0.00$. **P2 holds as a health gate:** all 3,000 ∞ -budget pairs mirrored integer-exactly (set counts and step counts compared as integers, never floats), 0 deviations, duplicate-mean set-share 0.5000 — the run-scale echo of the Phase 1 equivalence anchor.

Two structural readings. The curve is concave with an elbow between 16 and 48 bits: the first 16 bits buy 0.26 of set-share, the next 32 buy 0.10, everything past 64 buys 0.0023. And the axis saturates early: at 128 bits every pair already mirrors exactly and the per-seed set counts match the ∞ cell’s (Δ and SE both zero) — not because the games are identical (the two cells’ mean lengths differ by about 0.002 moves), but because no retained-versus-evicted difference above 128 bits ever moved a set on these 3,000 deals. The worst-case pool is hundreds of bits (Section 4); the *operative* pool, under this eviction policy and these deals, fits in 128. Bounded games are also longer: mean engine moves run from 683.2 at ∞ up to 798.2 at the floor, about 17% longer — memory-poor tables take longer to finish, a fact that returns in Section 7.4.

Table 2: E2, tier calibration: each shipped tier vs. the same reference opposition on the head of the ladder seed list, 3,000 pairs each, interpolated onto the E1 curve [14].

Tier	Set-share	SE	Bits-equivalent
easy	0.0446	0.0011	0 [0, 0] — clamped: below the 0-bit floor
medium	0.4656	0.0024	32.2 [30.8, 35.1]
hard	0.5006	0.0007	none finite — above every finite rung

7.2 P3: where the shipped tiers land

Table 2 is the measurement the ladder exists to provide, and the verdict is **mixed**, exactly as the rule reads it: the ordering easy < medium < hard holds on point estimates, but not every tier is finitely placeable.

The medium tier is worth 32.2 bits [30.8, 35.1] — a genuinely useful number: the full knowledge engine without constraint-refined hit probability plays like a contestability-ranked memory of about sixteen card facts. The hard tier’s share, 0.5006 ± 0.0007 , sits above every finite rung (its interval maps to $[85.5, \infty)$ in bits); it is not the mirror of the reference — the nonzero paired SE says the two disagree on some deals — but no finite budget reproduces it.

The easy tier is the headline of this table: 0.0446 ± 0.0011 , *below the ladder’s own zero-bit floor* of 0.1313 ± 0.0017 . The bits-equivalent is clamped at zero with the artifact’s note printed as measured: the share sits at or below the 0-bit floor. Read plainly: a bot that remembers *nothing* but reasons soundly over its own hand and the public board state, and never blunders, is worth nearly three times the set-share of a bot that remembers six events and dices away a quarter of its asks. The noise is not a memory model; it is a tax collected at random, and Section 7.3 shows its shape is wrong too.

7.3 P4–P6: the evidence-age curves

E3 asks the S45 question of every policy in the E1/E2 records: when a certain ask is available — a card whose holder the public log established at a known past event — does the policy take it, and does the answer depend on how old the evidence is? Table 3 reports the committed statistics per policy.

P5 — mixed, with the predicted structure where the model binds. Every budget from 0 through 48 bits decays significantly, and the half-lives, where defined, are 5/13/17/33/49 events at 0/8/16/24/32 bits — strictly increasing in bits, the S45 signature arriving exactly as the construction predicts: a bigger budget keeps old facts alive longer. But the registered rule demanded decay *and* a defined half-life at *every* finite budget, and from 48 bits up no band meeting the registered 200-observation floor falls to half the youngest-band rate (48 bits still decays at $z = 3.22$ but earns no half-life; 64–128 bits are statistically flat). The verdict prints as mixed: every budget decays — no; half-life defined everywhere and non-decreasing with a strict increase — no.

P4 — mixed, and instructive about estimators. The ∞ -budget bounded policy is age-flat under the registered rule (-0.0145 ± 0.0129 , $z = -1.13$ over 470 seeds). The unbounded reference is *not*: -0.0212 ± 0.0049 , $z = -4.28$ over 2,571 seeds — a small but significant tilt in the *anti*-decay direction (within a seed, old-evidence certain asks are exploited slightly more often than young ones). One of two full-memory policies flat is, by the fixed rule, mixed. The instructive part is the **composition effect** the artifact makes visible: pooled, the reference’s young and old rates are 0.5840 and 0.4561 — an apparent 13-point decay — while the seed-clustered estimator, comparing within games, finds the opposite sign. Both numbers are printed because they measure different things:

Table 3: E3, evidence age: exploit rate of available certain asks, young band (ages 1–8) and old band (33+), with the seed-clustered decay (young – old within seed) and the half-life where defined [14]. Pooled and clustered estimators can disagree (see text).

Policy	Young rate	Old rate	Clustered decay $\pm$ SE	z	Half-life
bounded 0	0.1030	0.0186	$+0.0661 \pm 0.0026$	25.22	5
bounded 8	0.2798	0.0574	$+0.1611 \pm 0.0052$	30.72	13
bounded 16	0.3626	0.1154	$+0.1055 \pm 0.0076$	13.92	17
bounded 24	0.4553	0.1646	$+0.0891 \pm 0.0104$	8.55	33
bounded 32	0.5060	0.2632	$+0.0572 \pm 0.0119$	4.81	49
bounded 48	0.5491	0.3626	$+0.0422 \pm 0.0131$	3.22	—
bounded 64	0.5576	0.4091	-0.0044 ± 0.0130	-0.34	—
bounded 96	0.5595	0.4244	-0.0145 ± 0.0129	-1.13	—
bounded 128	0.5595	0.4244	-0.0145 ± 0.0129	-1.13	—
bounded ∞	0.5595	0.4244	-0.0145 ± 0.0129	-1.13	—
reference (unbounded)	0.5840	0.4561	-0.0212 ± 0.0049	-4.28	—
old easy tier	0.4394	0.0266	$+0.4115 \pm 0.0024$	170.20	7

Table 4: E3, the old easy tier’s exploit rate and hit rate by evidence age [14]. The memory window is 6 events; the cliff between the 5–6 and 7–8 bands is the window’s edge, seen from outside.

Age band	Available	Exploit rate	Hit rate
1–2	34,816	0.5097	0.8015
3–4	46,943	0.4936	0.8067
5–6	25,064	0.5265	0.8084
7–8	17,138	0.0205	0.5557
9–12	30,822	0.0359	0.4717
13–16	24,370	0.0300	0.4728
17–24	37,003	0.0279	0.4702
25–32	25,595	0.0251	0.4703
33–48	30,450	0.0252	0.4611
49–64	14,319	0.0260	0.4548

games in which 33+-aged certain asks are still *available* at all are systematically unusual games (long, contested, with located cards left unconsumed), so the pooled old band conflates position quality with age. The clustered estimator is the verdict’s input for exactly this reason, and wherever the two disagree in this paper, both are shown.

P6 — mixed, and the motivating contrast lands. Table 4 is what “25% noise plus a window” looks like as a memory model. Inside the six-event window the exploit rate holds near 0.51; at age seven it falls off a cliff to 0.02 and stays there for the rest of the game. The registered cliff statistic is $+0.4861 \pm 0.0019$, $z = 258.87$ — for scale, the largest cliff any bounded budget shows at the same boundary is 0.1215 ± 0.0033 (8 bits), and the unbounded reference’s is 0.0718 ± 0.0027 . But the tier is *not* flat inside its window: the ages 1–3 versus 4–6 split is -0.0360 ± 0.0035 ($z = -10.31$) — the rate *rises* slightly with age inside the window — so one of the two registered conditions fails and the verdict prints mixed. The substantive contrast survives the mixed verdict intact: human memory, and the bounded ladder built to imitate it, decays smoothly with half-lives that scale with capacity; the noise tier is a step function — perfect recall to a hard edge, near-total amnesia beyond it — and no human plays like that [8, 2].

Table 5: E4: v1.0 classifier end-of-game top-1 over ecologies with both teams bounded, 36 pairings $\times$ 50 games $\times$ 6 seats per cell [14]. Chance is $1/9 \approx 0.111$; the committed full-strength baseline is 0.2243 [13]. Δ is the per-seed paired adjacent delta; rule $\Delta \geq -2 \cdot \text{SE}$.

Bits	Seats	Top-1	Δ from prev.	z
16	10,800	0.1542	—	—
32	10,800	0.2101	$+0.0559 \pm 0.0073$	7.70
64	10,800	0.2295	$+0.0194 \pm 0.0050$	3.87
∞	10,800	0.2243	-0.0053 ± 0.0019	-2.75

Table 6: E4 per-style end-of-game top-1 by budget [14]. Descriptive — the artifact commits no per-style SEs; 1,200 seats per entry. The ∞ column is the v1.0 baseline’s per-style accuracy, reproduced exactly.

True style	16 bits	32 bits	64 bits	∞
Balanced	0.0158	0.0433	0.0600	0.0558
Blitz	0.0592	0.0758	0.0925	0.0925
Punter	0.2742	0.1717	0.1333	0.1292
Banker	0.1075	0.1375	0.1250	0.1258
Turtle	0.2733	0.3525	0.3142	0.2717
Hoarder	0.1825	0.2575	0.3125	0.3092
Scout	0.2667	0.3675	0.4133	0.4125
Ghost	0.1608	0.3775	0.4983	0.5042
Archivist	0.0475	0.1075	0.1167	0.1175

7.4 P7 refuted: memory pressure makes styles *more* classifiable

P7 predicted that memory pressure erodes the behavioural signature: less memory, less characteristic play, lower classifier accuracy. Table 5 refutes it at both ends. At the bottom of the grid the direction is as predicted — 16-bit ecologies read at 15.4%, well below the 22.4% full-strength baseline. But the top of the grid inverts: *64-bit ecologies are more classifiable than full-strength ones*, 23.0% against 22.4%, and the registered rung test rejects — the $64 \rightarrow \infty$ delta is -0.0053 ± 0.0019 , $z = -2.75$, past the $\Delta \geq -2 \cdot \text{SE}$ bound. One violating rung refutes: **P7 is refuted**. The review’s multiplicity annotation does not soften it: over the three-rung family, the violating rung’s one-sided $p = 0.0029$ becomes $p = 0.00885$ under Bonferroni $\times 3$ — still a violation at the corrected level. The ∞ cell reproduces the committed v1.0 baseline to the last digit (0.22425925... in both artifacts) — the same games, by the anchor pin — so the comparison against v1.0 is an identity, not a replication.

Where does the extra classifiability come from? Table 6 (descriptive — no per-style SEs are committed) shows the readings redistribute rather than uniformly rise: under deep pressure the Ghost, v1.0’s one recognisable style, collapses (0.5042 at ∞ to 0.1608 at 16 bits) while the Punter more than doubles (0.1292 to 0.2742) — bounded ecologies push behaviour into regions where different fingerprints are nearest. Two mechanisms plausibly combine, and E4 cannot separate them: the read seat’s own behaviour changes, and the *ecology* changes — with all six seats bounded, opponents miss more, games run longer (Section 7.1 measured 17% on the one-sided ladder), and every length-sensitive feature shifts against fingerprints calibrated on full-strength play [12, 6]. Which of the two carries the P7 shift is precisely what E4 cannot say — the Phase 2 review’s finding, and the reason E4b exists.

Table 7: E4b: single-seat top-1 at the bounded read seat, pilot (50 seeds, 1,800 reads per cell) and power run (300 seeds, 10,800 reads per cell; the P8 verdict of record) [14]. Cell SEs are seed-clustered; Δ per-seed paired; rule $\Delta \geq -2 \cdot \text{SE}$.

Bits	Pilot (1,800 reads)		Power run (10,800 reads)	
	Top-1	Δ from prev.	Top-1	Δ from prev.
16	0.1678 ± 0.0100	—	0.1622 ± 0.0040	—
32	0.1678 ± 0.0104	$+0.0000 \pm 0.0118$	0.1698 ± 0.0044	$+0.0076 \pm 0.0046$
64	0.1700 ± 0.0105	$+0.0022 \pm 0.0066$	0.1642 ± 0.0043	-0.0056 ± 0.0031
∞	0.1683 ± 0.0102	-0.0017 ± 0.0026	0.1634 ± 0.0043	-0.0007 ± 0.0010

7.5 P8: the single-seat design, at pilot and at power

Table 7 carries both E4b runs, pilot and power, side by side — both reported whatever they say, as registered. In the power run — the P8 verdict of record — the single-seat curve is flat: 16.22%, 16.98%, 16.42%, 16.34% across the grid, no adjacent rung violating the rule (the most negative, 32→64 at -0.0056 ± 0.0031 , $z = -1.83$, sits inside the 2-SE bound; its Bonferroni-corrected one-sided p is 0.1018). **P8 is confirmed**, now at 99.97% post-hoc power at the P7 effect size (MDE 0.0019 under the registered rule) rather than the pilot’s 52.4% (MDE 0.0051). The ∞ health gate held in both runs: every ∞ -budget game — 1,800 in the pilot, 10,800 in the power run — reproduced the all-bare v1.0 harness game event-for-event, read-for-read and step-for-step, 0 deviations.

Two features of the single-seat numbers deserve explicit note. First, the absolute level: a single bounded seat among five full-strength ones reads at about 16–17% at every budget. That level is not comparable to E4’s cells or to the v1.0 baseline: P8’s reads are always the pairing’s *first* style, so the triangular pairing scheme weights the read distribution toward the early roster (2,400 Balanced reads down to 300 Ghost, and the Archivist is never the read truth), identically at every budget. The within-column comparisons P8 is built on are unaffected; the level is not the v1.0 baseline’s population and is not read against it. Second, the cross-design comparison the E4b-power registration required: at the violated rung (64→ ∞), P7’s delta is -0.0053 ± 0.0019 and P8’s is -0.0007 ± 0.0010 ; the difference of deltas is $+0.0045 \pm 0.0021$, $z = 2.11$, two-sided $p = 0.035$ — computed with independent errors, the power run’s seed list being disjoint from E4’s by construction. It is *labelled cross-design and enters no verdict*: P7’s rung lives in a world where both teams are bounded, P8’s in a world where only the read seat is, and a difference between the two designs is exactly what the attribution story predicts rather than a within-design test of anything.

7.6 The attribution, stated at its licensed strength

Assembled: E4 measures that bounding *everyone* at 64 bits makes end-of-game style reads better than full strength, refuting P7. E4b measures that bounding *only the read seat* produces a flat curve, at a power (99.97%) that would almost certainly have detected the P7-sized effect had it been present in that design. The licensed conclusion: within the designs measured, the P7 shift travels with the ecology — opponents, partners, game length, the event mix the features are computed over — and not with the read seat’s own end-of-game signature. What remains unlicensed, and is not claimed: that the read seat’s signature is unaffected in E4’s bounded-vs-bounded world (no design here isolates that), or any cross-design equality (the $z = 2.11$ difference is reported as a labelled observation). The pilot alone licensed none of this — its confidence interval contained the entire effect it was probing for — which is why the record distinguishes the pilot from the run of record, and why the correction of Section 6.4 was worth a commit.

8 Discussion

8.1 What the ladder buys

The replacement knob does what the old one could not, and the numbers are now attached. *Monotone*: nine of nine adjacent rungs non-negative under a paired test (Table 1). *Interpretable*: a tier is now a number — medium *is* 32.2 bits [30.8, 35.1] — and a handicap can be stated in units a player can reason about. *Human-shaped*: in the regime where the budget binds (0–32 bits), effectiveness decays with evidence age at half-lives 5–49 events that rise with capacity (Table 3) — the S45 curve, produced by construction rather than tuned. *Composable*: E4 ran all nine styles under budgets without any per-style work, because the arm restricts the knowledge, not the policy.

The concavity of the curve is itself a finding about **us54**: the first 16 bits — eight card facts, or a few facts plus a clutch of basis bits — carry about 71% of the distance from amnesia to full strength (0.26 of the 0.37 set-share gap), and everything past 64 bits is worth under a third of a point of set-share. Contestability-ranked memory is extraordinarily efficient at the bottom: what matters in this game is a small, current working set, not the archive.

8.2 The noise tier, priced and shaped

The old easy tier now has two independent condemnations. Priced on the ladder, it is worth less than zero bits — the memoryless-but-sound floor beats it by nearly a factor of three in set-share (Table 2), because a 25% uniform blunder rate and a guessing declarer destroy more value than a six-event memory saves. And shaped on the evidence-age axis, it is a step function with a $z = 259$ cliff at its window edge (Table 4) where humans — and the bounded ladder — show smooth decay. A learner watching the noise tier sees perfect play interleaved with inexplicable blunders; a learner watching a 16-bit bot sees something recognisable: a player who follows the current fight closely and forgets what happened twenty asks ago. The replacement is not merely stronger-per-unit-of-difficulty; it fails the way the thing it simulates fails.

8.3 What P7’s refutation means, and does not mean

The refutation is genuinely informative about the observation problem. The v1.0 instrument [6] was calibrated on full-strength play, and its accuracy was bounded by how rarely styles diverge on identical views [7, 10]. Bounded ecologies move play into regions the fingerprints never saw — longer games, more misses, different feature distributions — and the reads *redistribute*: the classifier is not recovering more style signal at 64 bits so much as reading a shifted population against fixed references, and the shift happens to raise top-1. That interpretation is consistent with the E4b result: hold the population at full strength except for one seat, and the effect vanishes at high power. It is also a caution for any deployed read of handicapped play: a classifier calibrated at full strength does not degrade gracefully under ecology shift — it changes its answers in structured ways (the Ghost collapse and Punter surge of Table 6) that look like signal.

What the refutation does not support: any claim that memory pressure leaves individual signatures intact in general. P8’s flatness is a statement at end-of-game reads, over this roster, against these fingerprints, in a design where five of six seats hold the ecology fixed. The v1.5 measurements say nothing about mid-game reads, cross-budget fingerprints, or classifiers recalibrated on bounded play — all of which are the natural next instruments if bounded opponents ship at the play table.

9 Threats to validity

1. **The ladder is an operationalisation, not a canon.** “Bits” are the framework’s accounting units under one tariff, one retirement rule, one subsumption rule, and one eviction policy; they are not Shannon information (a one-bit basis fact can carry more decision value than a two-bit card fact — the tariff prices storage, not worth). A different defensible spotlight scoring or tie order would produce a different — presumably similar, but unmeasured — curve. Bits-equivalents (Table 2) are relative to *this* ladder.
2. **E4’s both-teams design, and what E4b licenses.** P7’s refutation is a property of bounded-vs-bounded ecologies read by a full-strength-calibrated classifier; E4b isolates the read seat only within its own single-seat design, at end-of-game reads, at its measured power. The cross-design difference ($z = 2.11$) is an observation, not a test; no design here measures the read seat’s signature *inside* a bounded ecology.
3. **Fingerprints are calibrated on full-strength play.** Every E4/E4b read uses the committed v1.0 fingerprints [12]. The P7 movement therefore conflates behavioural change with calibration mismatch by construction; that conflation is the finding’s substance in Section 8.3, but it caps what any accuracy number here says about intrinsic distinguishability.
4. **The E3 estimator is itself an operationalisation.** “Certain asks” are defined through public hits only; availability is conditional on the position, and the composition effect of Section 7.3 shows how strongly availability selects on game character — hence the seed-clustered estimator for every verdict, with pooled rates printed beside it. Half-lives require 200-observation bands, so their absence at 48+ bits partly reflects old-evidence availability thinning (strong players consume certain asks young), not only flatness.
5. **Retained logs elide two derivable payloads.** The E1/E2 records keep the full public log event-for-event, but a claim’s assignments and true holders, and the final score snapshot, are elided as derivable or unused by the E3 estimator [14]. A future estimator needing them would need a re-run, not a re-read.
6. **Single machine, single implementation, closed population.** One engine, one rule set (pinned by the artifact’s rules hash), self-play within one roster; the usual transfer caveats [5] apply in full. The equivalence anchors and mirror gates certify harness symmetry and the bounded arm’s consistency with the unbounded engine — not the engine’s own correctness, which rests on its audits [4, 15].
7. **The E4 grid is coarse.** Four budgets, and the P7 violation is one rung; the shape between 64 bits and ∞ is uninstrumented. The Bonferroni annotation addresses the family actually tested, not the grid that might have been.

10 Conclusion

FishAI v1.5 set out to replace a difficulty knob that lied — noise dressed as weakness — with one that tells the truth, and the measurements say it does: a bit-budget memory with contestability eviction is monotone at every measured rung, saturates by 128 bits, prices the shipped tiers (medium = 32.2 bits; the old easy tier below the zero-bit floor it was meant to sit above), and fails the way people fail, with decay half-lives that grow with capacity. The pre-registered suite kept the project honest twice over: P7’s refutation — mild memory pressure makes whole ecologies *more* classifiable,

not less — survived its multiplicity annotation and became the paper’s most interesting result; and the E4b pilot’s overclaim was caught by its own review, corrected on the record, and settled by a 300-seed run of record at matched read count, which localised the effect, within the designs measured, in the ecology rather than the seat. The correction sequence is reported in full because it is the method working as designed: registrations before runs, reviews after them, and conclusions cut back to what the data licence — whichever direction that points.

Reproducibility. The engine is deterministic; the committed artifact [14] carries the configuration, seed prefixes, records digests, engine commits, rules hash, the verbatim predictions and all verdicts, with the E4b pilot retained beside the run of record; every number here is a field of it or of the committed v1.0 artifact [13], and the suite re-runs from `npm run bounded` [15].

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